V.L Memorial Public School

475 Vijay Nagar

CHAPTER-7-QUESTIONS-CLASS8TH-SCIENCE

Class 08 - Science (Curiosity)

Section A

1. How do liquid particles differ from solid particles? **[1]**

   a) Liquid particles remain fixed
   b) Liquid particles move freely
   c) Liquid particles cannot move
   d) Liquid particles escape immediately

2. What happens to particles when a solid reaches its melting point? **[1]**

   a) They move away from fixed positions
   b) They become completely stationary
   c) They lose all thermal energy
   d) They become tightly packed together

3. Why can matter particles not be seen through an ordinary microscope? **[1]**

   a) They are extremely distant
   b) They are extremely small
   c) They are extremely heavy
   d) They are extremely colourful

4. How does the incompressibility of liquids compare to gases? **[1]**

   a) Liquids are not compressible but gases are
   b) Liquids are less compressible than gases
   c) Liquids are more compressible than gases
   d) Liquids are equally compressible as gases

5. How does sugar behave when dissolved in water? **[1]**

   a) Changes its appearance but not its taste
   b) Remains visible but changes its state
   c) Settles at the bottom of the glass
   d) Breaks into constituent particles

6. Why gases and liquids are both classified as fluids. **[1]**

   a) Because they cannot be seen easily
   b) They occupy any container
   c) Both have fixed shapes
   d) Both can flow and do not have a fixed shape

7. Why does adding sand to water increase the total volume? **[1]**

   a) Sand dissolves between particles
   b) Sand changes into water
   c) Sand removes water particles
   d) Sand occupies space in water

8. How does temperature affect the movement of particles in water? **[1]**

a) The movement stops completely at room temperature
b) Temperature has no effect
c) Particles move faster in hot water and slow in cold
d) It slows down in hot water

9. When two gases are mixed, they: **[1]**

a) Do not mix
b) React to form liquid
c) Diffuse into each other
d) Settle at the bottom

10. Why do sugar particles seem to disappear in water? **[1]**

a) They become invisible due to dilution
b) They chemically transform into another substance
c) They dissolve and occupy interparticle spaces
d) They become microscopic elements

11. What happens to rock pieces as rivers continue flowing through mountainous regions? **[1]**

a) They break down into finer materials
b) They become larger solid rocks
c) They change directly into sugar
d) They dissolve completely into water

12. What happens to interparticle attractions when a liquid is heated toward its boiling point? **[1]**

a) They remain unchanged
b) They become stronger
c) They become weaker
d) They become permanent

13. If a rock is broken into sand, clay, and pebbles, how is this change best described? **[1]**

a) A transformation into new substances
b) An irreversible chemical reaction
c) A physical reduction of size
d) A change in the state of matter

14. In which scenario will potassium permanganate spread the fastest? **[1]**

a) In hot water
b) In dry air
c) In ice-cold water
d) In a closed container

15. What can be inferred about the size of constituent particles? **[1]**

a) The same size as visible particles
b) Larger than the basic unit of matter
c) Smaller than microscopic level
d) Comparable to the size of each grain in sand

16. In which condition does a solid occupy more space when added to water? **[1]**

a) When it is insoluble
b) When it forms bubbles
c) When it dissolves
d) When it evaporates

17. How can particles in solids gain enough energy to become liquid? **[1]**

a) Compressing the solid tightly
b) Increasing thermal energy to overcome attraction
c) By applying cold water
d) Shaking the particles vigorously

18. **Assertion (A):** The process of grinding chalk results in a chemical change. **[1]**
Reason (R): In a chemical change, new substances with different properties are formed.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

19. **Assertion (A):** Particles in a solid vibrate but do not move from their positions. **[1]**
Reason (R): The interparticle forces in solids are weak.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

20. **Assertion (A):** Sugar dissolves in water because its particles break up into constituent particles. **[1]**
Reason (R): The constituent particles of sugar are small enough to occupy interparticle spaces in water.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

21. **Assertion (A):** Interparticle spaces are present between particles of matter. **[1]**
Reason (R): Matter is composed of tiny particles that fill the available spaces when mixed.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** At room temperature, urea remains solid. **[1]**
Reason (R): Urea has a higher melting point than ice.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Solids have a definite shape whereas liquids do not. **[1]**
Reason (R): The interparticle forces are stronger in solids than in liquids.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Gases can be compressed easily as compared to solids and liquids. **[1]**

Reason (R): The particles in gases are closely packed with very little inter-particle space.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. Who first proposed the idea of Parmanu? **[1]**

26. Why do sand particles not dissolve in water? **[1]**

27. How does the volume of a gas change when compressed in a syringe? **[1]**

28. What are the spaces between particles called? **[1]**

29. What does the particulate nature of matter mean? **[1]**

30. What is the role of interparticle spacing in the properties of gases? **[1]**

31. What is meant by interparticle spacing in solids? **[1]**

32. Why are solids considered to have minimal interparticle spacing? **[1]**

Section B

33. A rubber band changes its shape when stretched. Can it be regarded as solid? **[2]**

34. Explain why potassium permanganate spreads faster in hot water than in cold water. **[2]**

35. Differentiate between boiling and evaporation. **[2]**

36. Explain how interparticle spacing affects liquids. **[2]**

37. What observable changes indicate the process of diffusion in liquids and gases? **[2]**

38. Describe why grinding chalk is considered a physical change rather than a chemical change. **[2]**

39. Explain how the movement of air particles helps fragrance from an incense stick spread throughout a room. **[2]**

40. Apply the concept of particle movement to explain why gases fill their containers uniformly. **[2]**

Section C

41. Differentiate the movement of particles in the solid, liquid, and gaseous states. **[3]**

42. Explain why a liquid has a definite volume but no fixed shape. **[3]**

43. Gases can be compressed, but solids cannot. Explain. **[3]**

44. What is the role of interparticle spacing in determining the properties of gases? **[3]**

45. How does the interparticle spacing influence the properties of liquids? **[3]**

46. Represent diagrammatically the changes in the arrangement of particles as ice melts and transforms into water vapour. **[3]**

47. Differentiate among solids, liquids, and gases based on **[3]**

    a. interparticle attraction,
    b. interparticle spacing,
    c. particle motion.

48. A solid is heated continuously. Explain how changes in particle motion and interparticle attractions lead to melting. **[3]**

Section D

49. **Read the following text carefully and answer the questions that follow:** **[4]**

Particles in matter are in constant motion, and their movement depends on thermal

energy. In solids, particles vibrate in place due to low thermal energy and strong interparticle forces. In liquids, particles move around within limited space as energy increases. In gases, particles have high energy and move freely in all directions, overcoming attractive forces. Thermal energy governs particle motion, spreading substances in liquids and gases, and influencing changes between states.

Questions:

a. Why do particles in gases move more freely than in solids? **(1)**

b. How does thermal energy affect particle motion in different states of matter? **(1)**

c. Compare particle movement and interparticle forces in solids, liquids, and gases. **(2)**

OR

d. Explain how particle motion is responsible for spreading substances in liquids and gases. **(2)**

50. **Read the following text carefully and answer the questions that follow:** **[4]**

The physical state of matter-solid, liquid, or gas-is determined by the strength of interparticle attractions. Solids have tightly packed particles with strong forces, giving them fixed shape and volume. Liquids have weaker attractions, allowing particles to move freely while retaining volume but no fixed shape. Gases have negligible interparticle forces, so particles move freely, occupying all available space. Temperature changes can alter particle motion, causing melting, boiling, or evaporation.

Questions:

a. Why do gases not have a fixed shape or volume? **(1)**

b. How do interparticle forces determine the state of matter? **(1)**

c. Compare the particle arrangement and movement in solids, liquids, and gases. **(2)**

OR

Explain how temperature affects the change of states of matter. **(2)**

51. **Read the following text carefully and answer the questions that follow:** **[4]**

Matter is made of extremely small particles called constituent particles. Large objects like rocks, sand grains, and sugar consist of countless tiny particles. These particles retain the properties of the substance and cannot be broken down further by physical means. When sugar dissolves in water, its particles spread out and occupy spaces between water molecules. Interparticle spaces explain why substances mix and why matter has volume but is composed of discrete particles.

Questions:

a. What are the smallest units of a substance that retain its properties called? **(1)**

b. Explain why sugar can dissolve in water but sand cannot, based on constituent particles and interparticle spaces. **(1)**

c. How do constituent particles support the idea that matter is composed of extremely small units? **(2)**

OR

Describe the role of interparticle spaces in the mixing of substances and in determining their physical state. **(2)**

Section E

52. Discuss the role of erosion in the formation of sand from larger rocks. **[5]**

53. Explain the interconversion of three states in terms of force of attraction and kinetic energy of the molecules. **[5]**

54. How is the high compressibility property of gas useful to us? **[5]**

55. The melting point of ice is 273.15 K. What does this mean? Explain in detail. **[5]**

56. What role does temperature play in changing the state of a substance from solid to liquid? **[5]**

57. Analyze how heating changes a solid into a liquid and a liquid into vapour. **[5]**

Section F

58. **State whether the given statement is True or False:** **[8]**

(a) Atoms and molecules are the smallest units of matter. **[1]**

(b) Water is not considered matter because it flows. **[1]**

(c) Liquids have a definite volume but no fixed shape. **[1]**

(d) The spaces between particles are the same in all states of matter. **[1]**

(e) Interparticle attractions decrease with an increase in distance. **[1]**

(f) Rivers transport sand and clay to plains by breaking down larger rocks along their course. **[1]**

(g) Pebbles and stones on a beach originate from the erosion of rocks in mountains. **[1]**

(h) The interparticle interactions in solids are very strong, and the interparticle spaces are very small. **[1]**

Section G

59. **Fill in the blanks:** **[10]**

(a) According to ancient philosophers matter consists of _____ constituents. **[1]**

(b) Thermal energy _____ when a solid changes to a liquid. **[1]**

(c) _____ effects how fast molecules moves. **[1]**

(d) The particles in a solid are packed very _____. **[1]**

(e) These tiny particles can be _____ or _____. **[1]**

(f) _____ is the change of gaseous state directly to solid state without going through liquid state, and vice-versa. **[1]**

(g) Grinding chalk into powder is an example of a _____ change. **[1]**

(h) Particles move the fastest in _____. **[1]**

(i) The minimum temperature at which a solid becomes a liquid is called its _____. **[1]**

(j) The attraction between particles is strongest in the _____ state. **[1]**